SAI VARDHAN PALLERLA

saivardhanpallerla40@gmail.com| +91-6309736136| github.com/Vardhansai18 | linkedin.com/in/vardhan18 | leetcode.com/u/saivardhanpallerla/

EDUCATION

VNR Vignana Jyothi Institute of Engineering & Technology **Hyderabad, India**

Bachelor of Technology in Electronics and communication Engineering *June 2019 – May 2023*

TECHNICAL SKILLS

Languages: C++, C, Python, Java, Perl, SQL, JavaScript
Systems Infrastructure: Linux, Docker, Kubernetes, Terraform, AWS, OpenStack, CI/CD
Networking & Protocols: TCP/IP, SNMP, IPSec, SSL/TLS, GRPC, REST APIs
Machine Learning & NLP: Lang Graph, Lang Chain, RAG Systems, Multi-Agent Workflows, Chroma DB, Fast API
Developer Tools & Frameworks: Git, Pytest, Playwright, Robot Framework, Postman, MySQL

EXPERIENCE

F5 Networks Inc ***Aug 2023 – Present***

Software Engineer – II *Hyderabad, India*

BIG-IP Deployment & SSL VPN Test Automation Framework

- Automated end-to-end provisioning of BIG-IP, client, and server environments on OpenStack using Terraform and Python, cutting setup time by **71% (3 hrs → 35 min)**, enabling rapid environment recreation for customer escalations and SRS.
- Re-architected enterprise regression test execution via concurrent pipeline orchestration and dynamic workload distribution, reducing runtime by **70% (**from **12 hours** to **3.5 hours**).
- Built an AI-powered Web Automation Agent integrating Model Context Protocol (MCP) with Playwright, enabling autonomous, session-aware multi-step browser interactions across Chromium, Firefox, and WebKit for complex end-to-end UI test workflows.
- Enhanced CI/CD pipeline automation by integrating Microsoft **Teams** webhooks following test execution to send daily status notifications, enabling remote monitoring on mobile devices without logging into the pipeline interface.
- **Migration:** Migrated and modernized a 10-year-old broken Jenkins test suite by upgrading Robot Framework syntax and dependencies, configuring dedicated execution agents, and resolving root infrastructure issues to restore full test automation.

Automated AI Assistant for Software Testing & Bug Fixing.

- Designed and engineered PRISM, an interactive multi-agent AI system that automates the end-to-end bug-fix lifecycle for BIG-IP network security defect orchestrating automated triage, source-level root cause analysis with confidence gating, and precise patch generation.
- Accelerated bug resolution timelines by **66%** (reducing defect lifecycle from 30 days to 10 days) across multiple cross-functional engineering teams, facilitating significantly faster release cadences while drastically reducing developer manual overhead.
- Designed and deployed an autonomous QA agent that parses feature requirement documents (PDRs), developer code, and REST APIs to generate end-to-end security, API, and UI test plans, slashing test planning cycles by **90%** (from 15 days to a single-day scan) across multiple F5 teams.

Distributed Network Management & SNMP Support for BIGIP

- Architected containerized microservice deployments for network management daemons on Kubernetes-based BIG-IP Next pods, integrating CI/CD test gates to cut pipeline build execution times by 55% and build failures by 45%.
- Engineered high-throughput SNMP agent infrastructure for BIG-IP Next, authoring an automated Python/Perl code-generation engine that compiled 1,500+ MIB object handlers and configuration templates across containerized runtimes.

Smart Interviews | Certificate ***Sept 2022 – April 2023***

Teaching Assistant – (Data Structures & Algorithms, Full Stack Development) *Hyderabad, India*

- Instructed **150+** undergraduate students in core Data Structures and Algorithms (DSA), conducting code reviews, space/time asymptotic complexity analysis, and algorithmic debugging in C++ and Java, Python.
- Mentored candidates through 1-on-1 technical interview coaching and problem-solving workshops, resulting in **a 90%** placement rate into competitive SWE internship and full-time roles.

PROJECTS

Tailor CV *JavaScript/TypeScript, Python, Fast API, Lang Graph, RAG, Chroma DB*

- Engineered a context-aware Chrome extension that enables students and working professionals to tailor resumes on the fly directly from job boards, boosting application-to-interview alignment and sustaining 90+ ATS compliance scores.
- Integrated client-side DOM scraping with an asynchronous backend powered by Fast API and Lang Graph, executing automated job-description parsing, vector retrieval via Chroma DB, and real-time prompt generation in sub-3 seconds.

ACHIEVEMENTS

- Code Chef 4-Star Coder: Ranked #48 globally in a worldwide algorithmic coding challenge among thousands of international competitors.
- Elected Chairperson, IEEE Student Branch (ECE): Spearheaded technical initiatives for student members, driving a 40% surge in active community participation and peer-led project development.